

Bagged Random Subset Methods for Forecast Combination: Exogenous Theory, Exact Bounds, and the Rate Window for k^*

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Abstract

This note collects the exogenous theoretical results for bagged Random Subset Methods (RSM) in forecast combination. The covariance matrix is treated as known. The endogenous case, Monte Carlo evidence, and empirical applications are deliberately omitted. Under the common-loading covariance structure, we derive the within-subset optimal weights, the exact infinite-bagged weights, a conservation law, and an exact expression for the bagged excess risk. We then derive an exact two-sided bracket for the excess risk $R_k^{\text{bag},\infty}$. The bracket gives a quadratic lower envelope and a linear upper envelope in $z_k = 1/k - 1/m$. Minimising the two envelope losses yields cube-root and square-root boundary rates. To interpret these boundary rates as a rate window for the true minimiser, we state an explicit marginal-envelope regularity assumption on $R_k - R_{k+1}$. Under that assumption, the optimal subset size satisfies

$$k_T^* = \Omega(T^{1/3}) \quad \text{and} \quad k_T^* = O(T^{1/2}),$$

provided the pool-size constraint does not bind. No power-law functional form for R_k is imposed.

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1 Scope and Main Message

The object of interest is the exogenous, known-covariance version of bagged RSM. A subset of size k is drawn from a pool of m forecasters; optimal weights are computed within the subset using the population covariance matrix; the bagged estimator averages over subset draws. The central theoretical question is how the population excess risk of the infinitely bagged estimator behaves as k increases, and what this implies for the optimal subset size once finite-sample inflation is added.

The key results are:

1. The finite-bagging bridge is exact:

$$Q_k^{\text{bag},G} = Q_k^{\text{bag},\infty} + \frac{1}{G} \left(Q_k^{\text{sub}} - Q_k^{\text{bag},\infty} \right).$$

2. Under the common-loading covariance structure, the infinite-bagged excess risk has an exact squared-distance representation:

$$R_k^{\text{bag},\infty} = \frac{1}{m^2} \sum_{i=1}^m \tau_i \left(g_i(k) - \frac{1}{\bar{\tau}} \right)^2.$$

3. This excess risk has an exact two-sided bracket:

$$\frac{(k\mathbb{E}_S[1/A_S] - 1/\bar{\tau})^2}{\sum_i \sigma_i^2 - m/\bar{\tau}} \leq R_k^{\text{bag},\infty} \leq \mathbb{E}_S[1/A_S] - \frac{1}{A_m}.$$

4. To leading order, the lower envelope is quadratic in $z_k = 1/k - 1/m$, while the upper envelope is linear in z_k .
5. Minimising the lower and upper envelope losses gives the boundary rates $T^{1/3}$ and $T^{1/2}$.
6. To turn these envelope rates into formal rate bounds for the true optimum, a marginal-envelope assumption on $R_k - R_{k+1}$ is sufficient. This assumption is weaker than imposing a power law $R_k = Cz_k^\alpha$.

2 Forecast-Combination Setup

Assumption 1 (Forecast errors). *There are m individual forecasts $f_t = (f_{1t}, \dots, f_{mt})'$ of a scalar target y_t . Forecast errors $u_t = f_t - y_t \iota$ satisfy*

$$\mathbb{E}(u_t) = 0, \quad \text{Var}(u_t) = \Sigma_e \succ 0, \quad \mathbb{E}(u_t u_{t'}') = 0 \quad \text{for } t \neq t'.$$

For any weight vector $w \in \mathbb{R}^m$ with $w' \iota = 1$, the combination forecast has loss

$$Q(w) = w' \Sigma_e w.$$

The scalar irreducible component s is common to all combinations and enters only through the final finite-sample loss.

The equal-weighted mean is

$$w^{\text{ave}} = \frac{1}{m} \iota, \quad Q^{\text{ave}} = \frac{1}{m^2} \iota' \Sigma_e \iota.$$

The infeasible population optimum is

$$w^{\text{opt}} = \frac{\Sigma_e^{-1} \iota}{\iota' \Sigma_e^{-1} \iota}, \quad Q^{\text{opt}} = (\iota' \Sigma_e^{-1} \iota)^{-1}.$$

3 RSM Definition and the Exact Bagging Bridge

Let $S \subset \{1, \dots, m\}$ be a uniformly drawn subset of size k . Let $\Sigma_{e,S}$ denote the $k \times k$ principal submatrix of Σ_e indexed by S . The population optimal weight within subset S , embedded in $\mathbb{R}^m$, is

$$v_S = P_S \frac{\Sigma_{e,S}^{-1} \iota}{\iota' \Sigma_{e,S}^{-1} \iota},$$

where P_S places the k computed weights in their original coordinates and sets all other coordinates to zero.

The average-subset risk is

$$Q_k^{\text{sub}} = \mathbb{E}_S[v_S' \Sigma_e v_S].$$

The finite-bagged weight based on G independent subset draws is

$$\hat{v}_{k,G} = \frac{1}{G} \sum_{g=1}^G v_{S_g},$$

and the infinite-bagged weight is

$$\bar{v}_k = \mathbb{E}_S[v_S].$$

The corresponding risks are

$$Q_k^{\text{bag},G} = \mathbb{E}[\hat{v}_{k,G}' \Sigma_e \hat{v}_{k,G}], \quad Q_k^{\text{bag},\infty} = \bar{v}_k' \Sigma_e \bar{v}_k.$$

Proposition 1 (Exact finite-bagging bridge). *For G independent subset draws,*

$$Q_k^{\text{bag},G} = Q_k^{\text{bag},\infty} + \frac{1}{G} (Q_k^{\text{sub}} - Q_k^{\text{bag},\infty}).$$

Hence

$$Q_k^{\text{bag},\infty} \leq Q_k^{\text{bag},G} \leq Q_k^{\text{sub}}.$$

Proof. Let $V = v_S$ and $\bar{v} = \mathbb{E}[V]$. Since $\hat{v}_{k,G} = G^{-1} \sum_g V_g$,

$$\mathbb{E}[\hat{v}_{k,G}' \Sigma_e \hat{v}_{k,G}] = \bar{v}' \Sigma_e \bar{v} + \mathbb{E}[(\hat{v}_{k,G} - \bar{v})' \Sigma_e (\hat{v}_{k,G} - \bar{v})].$$

Independence of the draws gives

$$\mathbb{E}[(\hat{v}_{k,G} - \bar{v})' \Sigma_e (\hat{v}_{k,G} - \bar{v})] = \frac{1}{G} \mathbb{E}[(V - \bar{v})' \Sigma_e (V - \bar{v})].$$

Finally,

$$\mathbb{E}[(V - \bar{v})' \Sigma_e (V - \bar{v})] = \mathbb{E}[V' \Sigma_e V] - \bar{v}' \Sigma_e \bar{v} = Q_k^{\text{sub}} - Q_k^{\text{bag},\infty}.$$

□

4 Common-Loading Structure

Assumption 2 (Common-loading covariance). *The forecast-error covariance matrix satisfies*

$$\Sigma_e = q\iota\iota' + D, \quad D = \text{diag}(\sigma_1^2, \dots, \sigma_m^2), \quad q \geq 0, \quad \sigma_i^2 > 0.$$

Define precisions

$$\tau_i = \sigma_i^{-2}, \quad A_S = \sum_{i \in S} \tau_i, \quad A_m = \sum_{i=1}^m \tau_i, \quad \bar{\tau} = \frac{A_m}{m}.$$

Under this structure, the Woodbury identity yields

$$\Sigma_{e,S}^{-1} = D_S^{-1} - \frac{q}{1 + qA_S} \tau_S \tau_S',$$

where $\tau_S = (\tau_i)_{i \in S}$. Therefore the constrained minimum-variance weight inside S is

$$w_{S,i} = \frac{\tau_i}{A_S}, \quad i \in S.$$

The common-loading parameter q does not affect the weights. The subset risk is

$$Q(S) = q + \frac{1}{A_S}.$$

The oracle and equal-weighted risks are

$$Q^{\text{opt}} = q + \frac{1}{A_m}, \quad Q^{\text{ave}} = q + \frac{1}{m^2} \sum_{i=1}^m \sigma_i^2.$$

It is useful to define the harmonic-mean variance scale

$$B := \frac{1}{\bar{\tau}} = \frac{m}{A_m}.$$

This is *not* the arithmetic mean of the variances. It is the harmonic mean of the idiosyncratic variances σ_i^2 . Let

$$\bar{\sigma}^2 := \frac{1}{m} \sum_{i=1}^m \sigma_i^2.$$

Define the arithmetic-harmonic heterogeneity gap

$$H_{AH} := \bar{\sigma}^2 \bar{\tau} \geq 1.$$

The inequality follows from the AM-HM inequality, with equality if and only if all σ_i^2 are equal. Then

$$Q^{\text{opt}} = q + \frac{B}{m}, \quad Q^{\text{ave}} = q + \frac{H_{AH} B}{m}, \quad Q^{\text{ave}} - Q^{\text{opt}} = \frac{B(H_{AH} - 1)}{m}.$$

Remark 1 (Two heterogeneity measures should not be conflated). *The object $H_{AH} = \bar{\sigma}^2 \bar{\tau}$ measures the arithmetic-vs-harmonic gap and appears in the exact bounds below. A different object sometimes used in simulations is the range ratio*

$$H_{\text{range}} := \frac{\max_i \sigma_i^2}{\min_i \sigma_i^2}.$$

These are not the same and should be denoted separately.

5 Infinite-Bagged Weights and Exact Excess Risk

For $i = 1, \dots, m$, define

$$g_i(k) := k \mathbb{E}_S \left[\frac{1}{A_S} \mid i \in S \right].$$

Then

$$\bar{v}_{k,i} = \mathbb{E}_S[v_{S,i}] = \frac{k}{m} \mathbb{E}_S \left[\frac{\tau_i}{A_S} \mid i \in S \right] = \frac{\tau_i g_i(k)}{m}.$$

Since $\sum_i \bar{v}_{k,i} = 1$, we have

$$\sum_{i=1}^m \tau_i g_i(k) = m.$$

Proposition 2 (Conservation law). *For every $k \in \{1, \dots, m\}$,*

$$\sum_{i=1}^m \tau_i \left(g_i(k) - \frac{1}{\tau_i} \right) = 0.$$

Equivalently, RSM redistributes weight relative to the equal-weighted benchmark without changing total mass.

Proof. From $\sum_i \bar{v}_{k,i} = 1$ and $\bar{v}_{k,i} = \tau_i g_i(k)/m$, we obtain $\sum_i \tau_i g_i(k) = m$. Also $\sum_i \tau_i (1/\tau_i) = m$. Subtracting gives the result. $\square$

Define

$$R_k^{\text{bag},\infty} := Q_k^{\text{bag},\infty} - Q^{\text{opt}}.$$

Proposition 3 (Exact excess-risk formula). *Under the common-loading structure,*

$$R_k^{\text{bag},\infty} = \frac{1}{m^2} \sum_{i=1}^m \tau_i \left(g_i(k) - \frac{1}{\bar{\tau}} \right)^2.$$

Proof. Let $\Delta_k = \bar{v}_k - w^{\text{opt}}$. Since both $\bar{v}_k$ and w^{opt} sum to one, $\iota' \Delta_k = 0$. Therefore the common-loading component vanishes:

$$\Delta_k' \Sigma_e \Delta_k = q(\iota' \Delta_k)^2 + \Delta_k' D \Delta_k = \sum_i \sigma_i^2 \Delta_{i,k}^2.$$

Moreover,

$$w_i^{\text{opt}} = \frac{\tau_i}{A_m} = \frac{\tau_i}{m\bar{\tau}}, \quad \bar{v}_{k,i} = \frac{\tau_i g_i(k)}{m}.$$

Hence

$$\Delta_{i,k} = \frac{\tau_i}{m} \left(g_i(k) - \frac{1}{\bar{\tau}} \right).$$

Substituting into $\sum_i \sigma_i^2 \Delta_{i,k}^2 = \sum_i \tau_i^{-1} \Delta_{i,k}^2$ gives the expression. $\square$

The boundary conditions are immediate. At $k = m$, $\bar{v}_k = w^{\text{opt}}$ and $R_m^{\text{bag},\infty} = 0$. At $k = 1$, $\bar{v}_1 = w^{\text{ave}}$, so

$$R_1^{\text{bag},\infty} = Q^{\text{ave}} - Q^{\text{opt}}.$$

6 Average-Subset Benchmark

The average-subset object is useful because it gives a transparent upper envelope for the bagged excess risk. Under the common-loading structure,

$$Q_k^{\text{sub}} = q + \mathbb{E}_S \left[\frac{1}{A_S} \right].$$

Lemma 1 (Taylor expansion of the expected inverse precision sum). *Let $\epsilon_S = A_S - k\bar{\tau}$. Then $\mathbb{E}_S[\epsilon_S] = 0$ and*

$$\text{Var}_S(A_S) = k \left(1 - \frac{k-1}{m-1} \right) \frac{1}{m} \sum_{i=1}^m (\tau_i - \bar{\tau})^2.$$

If $|\epsilon_S|/(k\bar{\tau})$ is uniformly small enough for the Taylor expansion to be controlled, then

$$\mathbb{E}_S \left[\frac{1}{A_S} \right] = \frac{1}{k\bar{\tau}} + \frac{\text{Var}_S(A_S)}{(k\bar{\tau})^3} + O \left(\frac{\mathbb{E}_S |\epsilon_S|^3}{(k\bar{\tau})^4} \right).$$

Proof. Write $A_S = k\bar{\tau} + \epsilon_S$. Expanding around $\epsilon_S = 0$ gives

$$\frac{1}{A_S} = \frac{1}{k\bar{\tau}} - \frac{\epsilon_S}{(k\bar{\tau})^2} + \frac{\epsilon_S^2}{(k\bar{\tau})^3} + O\left(\frac{|\epsilon_S|^3}{(k\bar{\tau})^4}\right).$$

Taking expectations and using $\mathbb{E}_S[\epsilon_S] = 0$ gives the claim. The variance formula is the standard finite-population formula for sampling without replacement. $\square$

Therefore, to leading order,

$$Q_k^{\text{sub}} - Q^{\text{opt}} = \mathbb{E}_S[1/A_S] - \frac{1}{A_m} \approx \frac{1}{\bar{\tau}} \left(\frac{1}{k} - \frac{1}{m} \right).$$

Define

$$z_k := \frac{1}{k} - \frac{1}{m}.$$

Then the average-subset excess risk is approximately linear in z_k .

If the finite-sample loss for the average-subset estimator is approximated by

$$L_k^{\text{sub}} \approx \left(A_0 + \frac{B}{k} \right) \frac{T-1}{T-k}, \quad A_0 = s + q, \quad B = 1/\bar{\tau},$$

then the first-order condition is

$$A_0 k^2 + 2Bk - BT = 0,$$

with positive root

$$k^{\text{sub},*} = \frac{-B + \sqrt{B^2 + A_0 BT}}{A_0} \sim \left(\frac{BT}{A_0} \right)^{1/2}.$$

Thus the average-subset benchmark produces a square-root rate.

7 Exact Two-Sided Bounds for the Bagged Excess Risk

The infinite-bagged excess risk cannot generally be written in closed form because it depends on the conditional harmonic means

$$\mathbb{E}_S \left[\frac{1}{A_S} \mid i \in S \right].$$

However, it can be bracketed exactly using only moment identities.

Define the normalised precision dispersion

$$V_\delta := \frac{1}{m\bar{\tau}^2} \sum_{i=1}^m (\tau_i - \bar{\tau})^2.$$

Also note that

$$\sum_{i=1}^m \sigma_i^2 - \frac{m}{\bar{\tau}} = m\bar{\sigma}^2 - \frac{m}{\bar{\tau}} = \frac{m(H_{AH} - 1)}{\bar{\tau}}.$$

This denominator is zero only in the homogeneous case.

Lemma 2 (Moment identities). *For every k ,*

$$\sum_{i=1}^m \tau_i g_i(k) = m, \quad \sum_{i=1}^m g_i(k) = km \mathbb{E}_S \left[\frac{1}{A_S} \right].$$

Proof. The first identity was shown above. For the second,

$$\sum_i g_i(k) = k \sum_i \mathbb{E}_S \left[\frac{1}{A_S} \mid i \in S \right] = k \sum_i \frac{m}{k} \mathbb{E}_S \left[\frac{1_{\{i \in S\}}}{A_S} \right] = m \mathbb{E}_S \left[\frac{k}{A_S} \right] = km \mathbb{E}_S \left[\frac{1}{A_S} \right].$$

□

Proposition 4 (Exact two-sided bracket). *Under the common-loading structure, for all $1 \leq k \leq m$,*

$$\boxed{\frac{(k\mathbb{E}_S[1/A_S] - 1/\bar{\tau})^2}{\sum_i \sigma_i^2 - m/\bar{\tau}} \leq R_k^{\text{bag},\infty} \leq \mathbb{E}_S[1/A_S] - \frac{1}{A_m}}$$

provided the denominator is positive. In the homogeneous case, both $Q^{\text{ave}} - Q^{\text{opt}}$ and $R_k^{\text{bag},\infty}$ vanish.

Proof. For the upper bound, Jensen's inequality gives

$$Q_k^{\text{bag},\infty} = (\mathbb{E}_S v_S)' \Sigma_e (\mathbb{E}_S v_S) \leq \mathbb{E}_S [v_S' \Sigma_e v_S] = Q_k^{\text{sub}}.$$

Subtracting $Q^{\text{opt}} = q + 1/A_m$ and using $Q_k^{\text{sub}} = q + \mathbb{E}_S[1/A_S]$ gives the ceiling.

For the lower bound, write

$$R_k^{\text{bag},\infty} = \frac{1}{m^2} \sum_i \tau_i g_i(k)^2 - \frac{1}{A_m}.$$

Minimise $\sum_i \tau_i g_i^2$ over all vectors $g = (g_1, \dots, g_m)$ satisfying the two identities

$$\sum_i \tau_i g_i = m, \quad \sum_i g_i = km \mathbb{E}_S[1/A_S].$$

Let $h_i = 1/\tau_i$ and write

$$g_i = \frac{1}{\bar{\tau}} + a \left(h_i - \frac{1}{\bar{\tau}} \right).$$

This automatically satisfies $\sum_i \tau_i g_i = m$. The second constraint pins down

$$a = \frac{m(k\mathbb{E}_S[1/A_S] - 1/\bar{\tau})}{\sum_i \sigma_i^2 - m/\bar{\tau}}.$$

The weighted norm of the residual direction is

$$\sum_i \tau_i \left(h_i - \frac{1}{\bar{\tau}} \right)^2 = \sum_i \sigma_i^2 - \frac{m}{\bar{\tau}}.$$

Substituting the minimum weighted norm into the exact excess-risk formula gives the lower bound. Since the true score vector satisfies the two identities, the minimum over the constrained class is a valid lower bound. □

7.1 Leading-Order Envelope

Using the Taylor expansion above,

$$k\mathbb{E}_S[1/A_S] - \frac{1}{\bar{\tau}} \approx \frac{\text{Var}_S(A_S)}{k^2 \bar{\tau}^3}.$$

For $1 \ll k \ll m$,

$$\text{Var}_S(A_S) \approx k \bar{\tau}^2 V_\delta,$$

so

$$k\mathbb{E}_S[1/A_S] - \frac{1}{\bar{\tau}} \approx \frac{V_\delta}{\bar{\tau}} \frac{1}{k}.$$

Equivalently, keeping the finite- m correction through $z_k = 1/k - 1/m$,

$$k\mathbb{E}_S[1/A_S] - \frac{1}{\bar{\tau}} \approx \frac{V_\delta}{\bar{\tau}} z_k.$$

Therefore the lower envelope is quadratic:

$$R_k^{\text{bag},\infty} \gtrsim C_L z_k^2, \quad C_L := \frac{V_\delta^2}{m(H_{AH} - 1)\bar{\tau}}.$$

The upper envelope is linear:

$$R_k^{\text{bag},\infty} \lesssim C_U z_k, \quad C_U := \frac{1}{\bar{\tau}} = B.$$

Hence the exact bracket gives the leading-order envelope

$$C_L z_k^2 \lesssim R_k^{\text{bag},\infty} \lesssim C_U z_k.$$

This does *not* impose a power law on R_k . It only says that the bagged excess risk is bracketed between a quadratic and a linear envelope.

8 Envelope Losses and Boundary Rates

Let

$$A := s + Q^{\text{opt}} > 0.$$

The exogenous finite-sample loss for the infinitely bagged estimator is approximated by

$$L_T(k) = (A + R_k) \frac{T-1}{T-k}, \quad R_k := R_k^{\text{bag},\infty}.$$

The two leading-order envelope losses are

$$L_L(k) = (A + C_L z_k^2) \frac{T-1}{T-k}, \quad L_U(k) = (A + C_U z_k) \frac{T-1}{T-k}.$$

For $k \ll m, T$, $z_k \approx 1/k$.

8.1 Lower Envelope: Cube-Root Boundary

Approximate

$$L_L(k) \approx \left(A + \frac{C_L}{k^2} \right) \frac{T-1}{T-k}.$$

The first-order condition gives

$$Ak^3 + 3C_L k - 2C_L T = 0.$$

Thus

$$k_L^* \sim \left(\frac{2C_L T}{A} \right)^{1/3}.$$

The quadratic lower envelope therefore generates the cube-root boundary rate.

8.2 Upper Envelope: Square-Root Boundary

Approximate

$$L_U(k) \approx \left(A + \frac{C_U}{k} \right) \frac{T-1}{T-k}.$$

The first-order condition gives

$$Ak^2 + 2C_U k - C_U T = 0,$$

with positive root

$$k_U^* = \frac{-C_U + \sqrt{C_U^2 + AC_U T}}{A} \sim \left(\frac{C_U T}{A} \right)^{1/2}.$$

The linear upper envelope therefore generates the square-root boundary rate.

Remark 2 (What the envelope minimisers do and do not prove). *The pointwise bounds imply*

$$L_L(k) \leq L_T(k) \leq L_U(k)$$

for each fixed k , up to leading-order approximation. Hence they bound the level of the loss curve. But pointwise bounds on a function do not automatically bound the exact location of its minimiser. The location of the minimiser depends on the marginal decline $R_k - R_{k+1}$. Therefore, to convert the envelope rates into formal rate bounds for the true minimiser, we need a marginal regularity assumption.

9 Marginal-Envelope Assumption and Rate Window for k^*

This section states the additional assumption needed to use the bounds for the rate of the true optimal subset size. The assumption is deliberately weaker than imposing

$$R_k = C z_k^\alpha.$$

It only controls the one-step decline of R_k .

Assumption 3 (Marginal envelope). *Let*

$$\Delta R_k := R_k - R_{k+1}.$$

For all relevant k with $1 \ll k \ll \min\{m, T\}$, suppose there exist constants $0 < d_L \leq d_U < \infty$, independent of T and k , such that

$$d_L (z_k^2 - z_{k+1}^2) \leq \Delta R_k \leq d_U (z_k - z_{k+1}).$$

Equivalently, since $z_k - z_{k+1} \asymp k^{-2}$ and $z_k^2 - z_{k+1}^2 \asymp k^{-3}$ for $k \ll m$,

$$d'_L k^{-3} \lesssim R_k - R_{k+1} \lesssim d'_U k^{-2}.$$

Remark 3 (Interpretation). *The level bracket says that R_k lies between quadratic and linear envelopes. The marginal-envelope assumption says that the one-step gain from increasing k is also between the marginal decline of those two envelopes. It rules out pathological sequences that satisfy the level bounds but decline through long flat plateaus or sudden jumps. It allows mixtures such as $R_k = a z_k^2 + b z_k$, slowly varying corrections, and smooth intermediate decay. It does not impose a power-law shape.*

We also need mild boundedness conditions.

Assumption 4 (Regularity of baseline and pool size). *As $T \rightarrow \infty$:*

- (i) The baseline satisfies $0 < a \leq A_T \leq \bar{A} < \infty$.
- (ii) The excess risk is uniformly bounded: $0 \leq R_k \leq \bar{R} < \infty$.
- (iii) The pool-size constraint does not bind at the square-root rate: $m_T/T^{1/2} \rightarrow \infty$. If this fails, the upper rate is truncated by m_T .

Theorem 1 (Rate window for the exogenous optimal subset size). *Let*

$$k_T^* \in \arg \min_{1 \leq k \leq m_T} L_T(k), \quad L_T(k) = (A_T + R_k) \frac{T-1}{T-k}.$$

Under the marginal-envelope and regularity assumptions,

$$k_T^* = \Omega(T^{1/3}) \quad \text{and} \quad k_T^* = O(T^{1/2}).$$

Equivalently, the optimal subset size lies in the cube-root to square-root rate window:

$$T^{1/3} \lesssim k_T^* \lesssim T^{1/2},$$

up to constants and subject to the pool-size constraint.

Proof. The loss decreases from k to $k+1$ if

$$L_T(k+1) < L_T(k).$$

Using the definition of $L_T(k)$ and cross-multiplying, this is equivalent to

$$(T-k)(R_k - R_{k+1}) > A_T + R_k.$$

Similarly, the loss increases from k to $k+1$ if the reverse inequality holds.

For the lower rate bound, use the lower marginal envelope. For $k \ll T$,

$$(T-k)(R_k - R_{k+1}) \gtrsim Tk^{-3}.$$

Because $A_T + R_k \leq \bar{A} + \bar{R}$, there exists a sufficiently small constant $c > 0$ such that for all $k \leq cT^{1/3}$,

$$(T-k)(R_k - R_{k+1}) > A_T + R_k.$$

Thus the loss is still decreasing throughout this region, so the minimiser cannot occur below the cube-root scale. Hence $k_T^* = \Omega(T^{1/3})$.

For the upper rate bound, use the upper marginal envelope:

$$(T-k)(R_k - R_{k+1}) \lesssim Tk^{-2}.$$

Because $A_T + R_k \geq a > 0$, there exists a sufficiently large constant $C < \infty$ such that for all $k \geq CT^{1/2}$ with $k \ll T$,

$$(T-k)(R_k - R_{k+1}) < A_T + R_k.$$

Therefore the loss is increasing beyond the square-root scale, so the minimiser cannot occur above it, except if the pool-size boundary binds. Hence $k_T^* = O(T^{1/2})$. $\square$

10 Recommended Theoretical Statement

The clean exogenous theoretical conclusion is:

Under the common-loading covariance structure and known Σ_e , the infinite-bagged RSM excess risk admits an exact two-sided bracket. To leading order, the lower envelope is quadratic in $z_k = 1/k - 1/m$, while the upper envelope is linear in z_k . Minimising the corresponding envelope losses gives cube-root and square-root boundary rates. Under a marginal-envelope regularity condition on $R_k - R_{k+1}$, the true exogenous optimum lies in the rate window

$$k_T^* = \Omega(T^{1/3}), \quad k_T^* = O(T^{1/2}).$$

No power-law functional form for R_k is required.

11 What Is Not Claimed

The results above do not claim that

$$R_k = Cz_k^\alpha$$

for some $\alpha \in [1, 2]$. That is an additional parametric approximation and is not needed for the bound-based theory.

The results also do not claim that the minimiser of the true loss is mechanically between the minimisers of the two pointwise envelope losses for every finite T . Pointwise bounds on loss levels do not by themselves imply pointwise bounds on argmins. The marginal-envelope assumption is the formal condition that makes the rate-window argument valid.

Finally, the endogenous case is not covered. Once Σ_e is estimated, the same sample covariance matrix is shared across subset draws, and the exogenous arguments need not carry over without additional assumptions.